FINMARK CORPORATION

From fragmented customer data to responsible predictive insight

ML Solution Presentation | Terminal Assessment

Team PETIX

Ma Carmela Paula Oreta • Srinder Kour Singh
Irisha Bea Montaño • Jasver Kour Singh

EVIDENCE FIRST. MODEL SECOND.

EDA • Data quality • Visual insights • Proposed ML path

EXECUTIVE SUMMARY

A useful foundation is in place; prediction is the next decision

Three source tables were cleaned and explored. The evidence supports a careful next step, not a performance claim.

8,942
EDA records
Across three tables; not unique people

3
Customer data sources
Demographics, transactions, social

7
Visualizations
Univariate, categorical, time, joined

0
Models evaluated
No target or model results yet

Customer profile

Observed ages span 18–70; mean 44.59 and median 45. Coverage is broad, but 362 ages in the EDA table were imputed.

Transactions

Named product categories have similar transaction counts. Amounts are widely spread; age has almost no linear association with transaction amount ($r \approx 0.032$).

Social & decision

Platform sentiment shares look broadly similar. Propose a time-aware purchase-propensity pilot only after target, date coverage, and customer linkage are approved.

Source: FinMark project data; Week 3 EDA and Week 7 visualization workflow.

OUR APPROACH

A traceable workflow from source data to a responsible ML proposal

We moved from understanding data quality, to describing customer behavior, to defining what would need to be true before prediction.

01 | PREPARE & AUDIT

Profile the three datasets, remove exact duplicates, preserve and flag conflicting IDs, assess data quality, and create documented analysis subsets.

02 | EXPLORE & VISUALIZE

Use summaries and seven charts to examine customer ages, transaction amounts, product mix, date coverage, platform sentiment, and joined customer behavior.

03 | PROPOSE & VALIDATE

Define a business outcome with stakeholders; verify time coverage and customer linkage; compare a leakage-safe baseline with candidate models before any pilot.

Gate before modeling: no approved target definition or verified date coverage means no valid predictive evaluation.

Source: FinMark project data; Week 3 EDA and Week 7 visualization workflow.

Data quality gates shape every conclusion

Each source began with 3,200 rows. Exact duplicates were removed; ID conflicts were retained for review, not silently resolved.

9,600
Raw rows
3,200 per source dataset

542
Exact duplicates
177 demographics · 185 transactions · 180 social

116
ID-conflict rows
46 demographics · 30 transactions · 40 social

8,942
Rows in EDA
2,977 + 2,985 + 2,980

Rules applied

- Preserve original values and add quality flags
- Treat missing categories as Unknown
- Impute screened ages with median 45 and flag
- Retain zero amounts; flag negative or invalid values
- Parse mixed dates day-first (source convention unconfirmed)

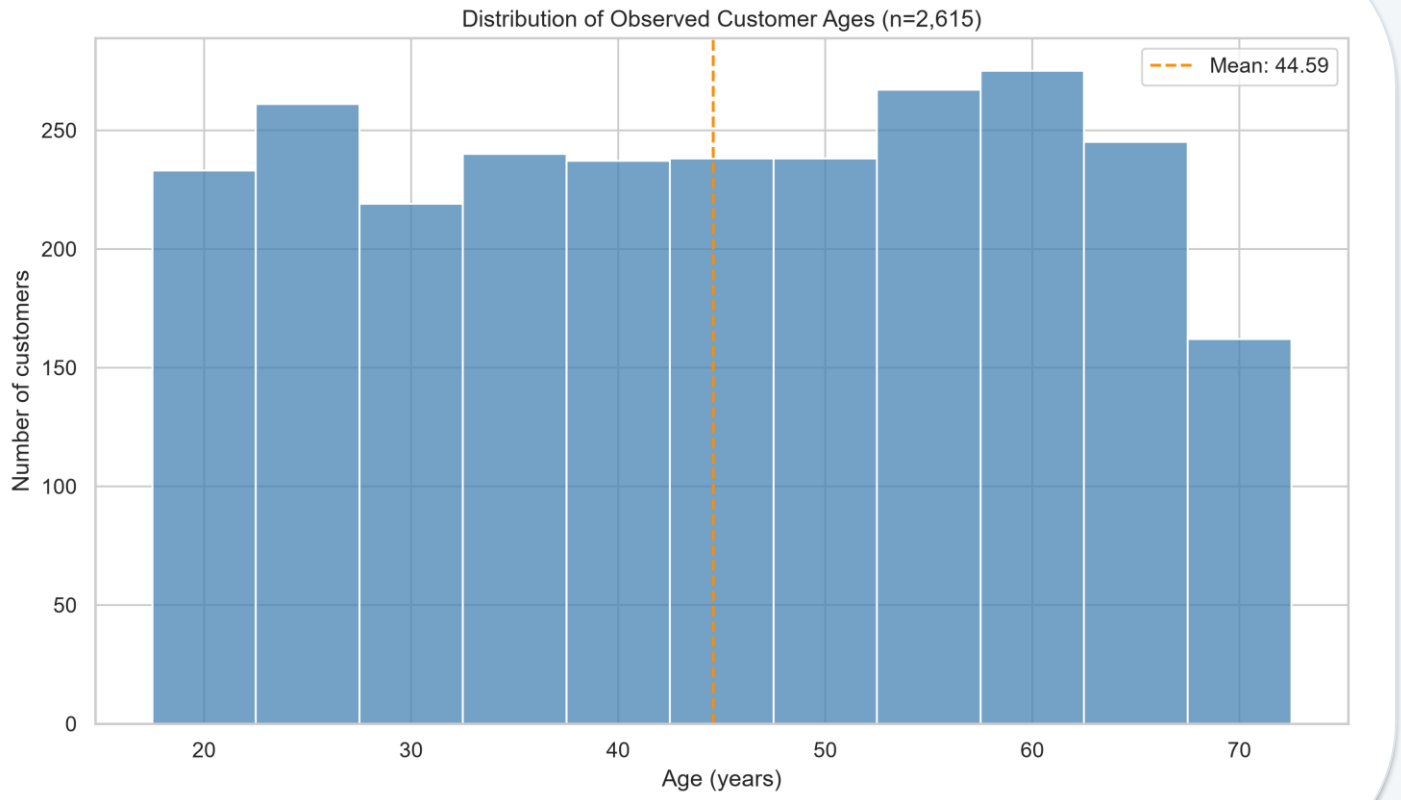
Limits to carry forward

- Amount analysis: 2,642 valid nonnegative values; 310 missing/nonnumeric and 33 negative excluded
- Conflicting IDs excluded from EDA; identity remains unresolved
- Unknown is missing information, not a true business category
- Date coverage and imputed ages need validation before modeling

Source: Week 3 README, audits, and preprocessing notebook. Counts are rows/records, not unique customers.

Observed ages are centered near 45

The chart excludes median-imputed ages and uses the 2,615 observed values.



WHAT THE DATA SHOWS

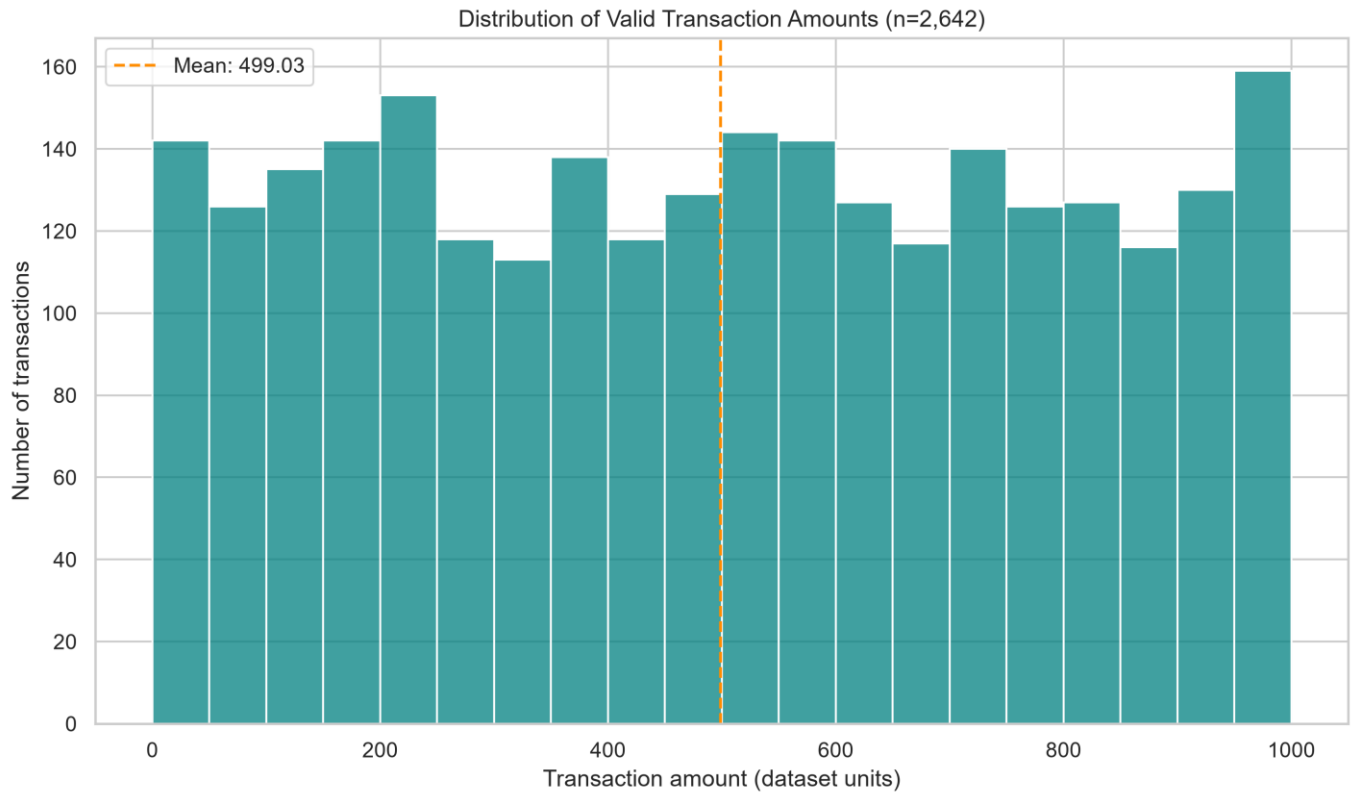
- Range: 18–70 years.
- Mean 44.59; median 45.
- The 58–62 bin is largest: 275 records (10.52%).
- The 68–72 bin includes observed ages only through 70.

Keep imputation flags; check age coverage and fairness before use.

Chart: 01_observed_age_distribution.png | Week 7 visualization workflow.

Transaction values are widely distributed

2,642 nonmissing, nonnegative amounts; zero-value transactions are retained.



WHAT THE DATA SHOWS

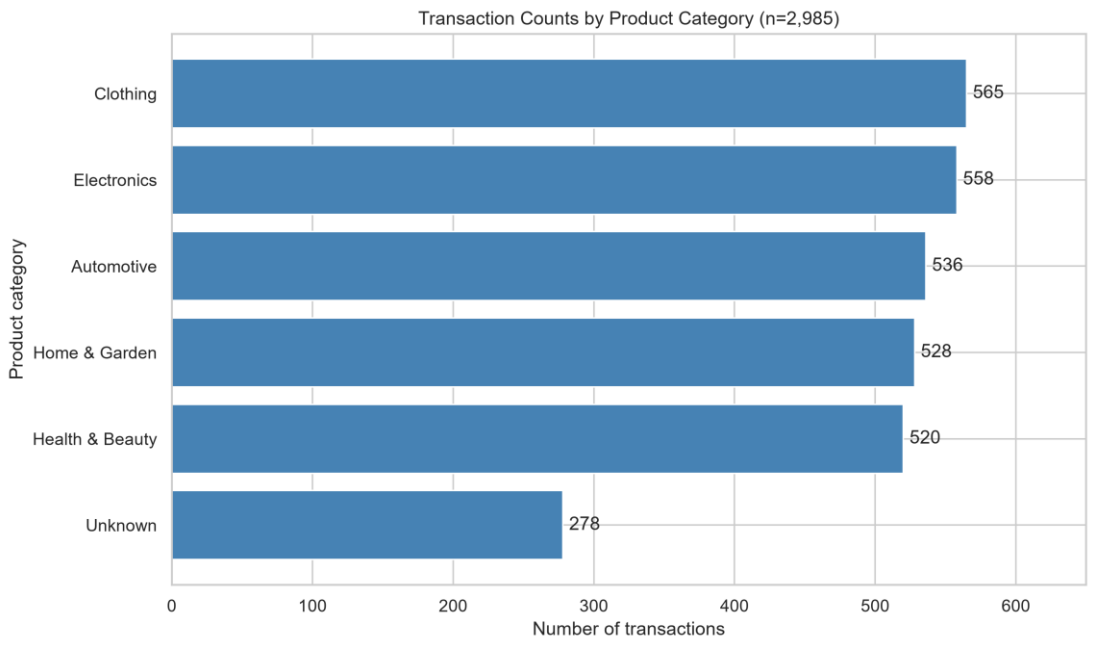
- Range: 0–999.86 dataset units.
- Mean 499.03; median 503.66.
- Sample standard deviation: 292.46.
- The similar mean and median do not imply values cluster near 500.

Scale only after defining the target and fitting transforms on training data.

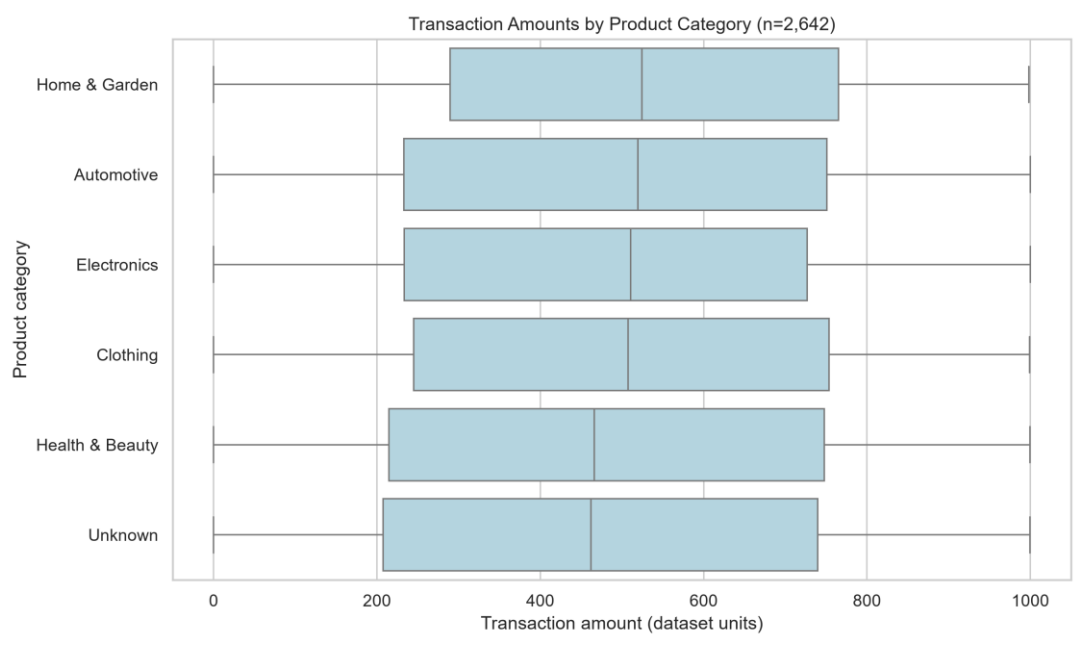
Chart: 02_transaction_amount_distribution.png | Week 7 visualization workflow.

Category mix is balanced; amount distributions overlap

Counts describe transaction frequency; the amount chart uses the valid-amount subset.



Counts across the five named categories range from 520 to 565.

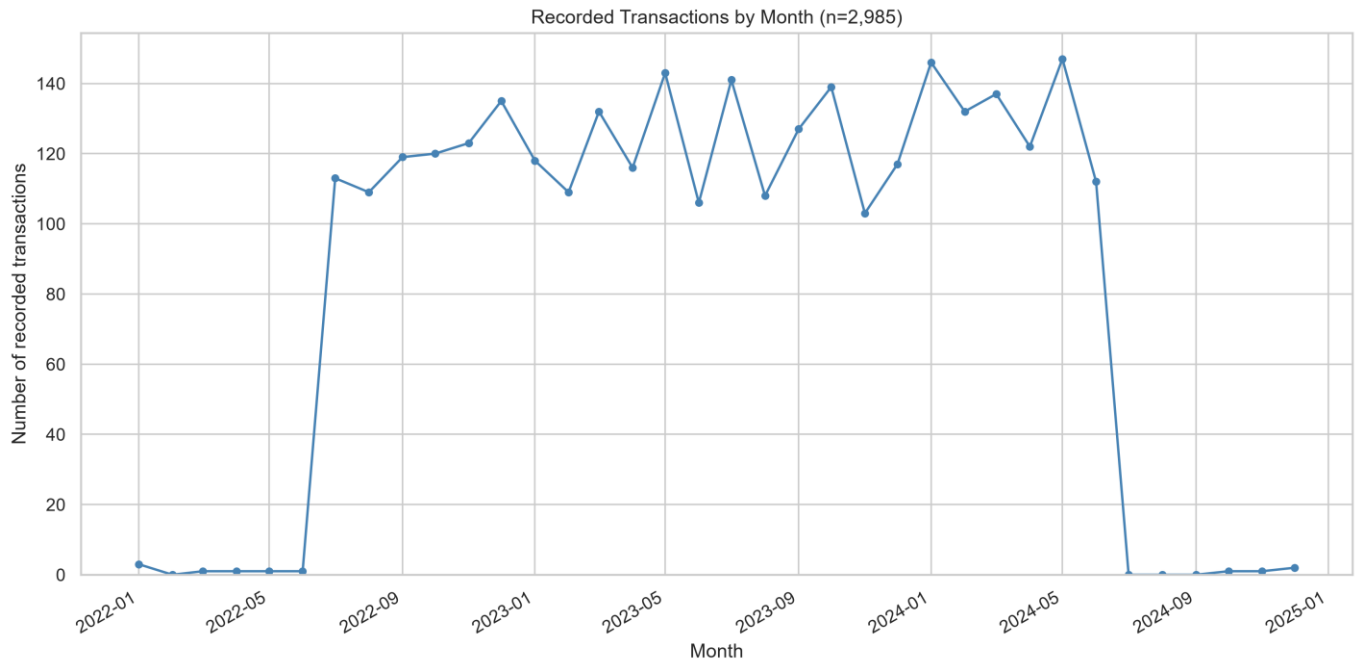


Home & Garden median: 524.62; Health & Beauty: 466.14.

Source: Week 7 charts; all category counts use 2,985 transaction records; amount comparison uses 2,642 valid values.

Recorded activity is concentrated in a central window

Dates span January 2022–December 2024, but coverage is uneven.



WHAT THE DATA SHOWS

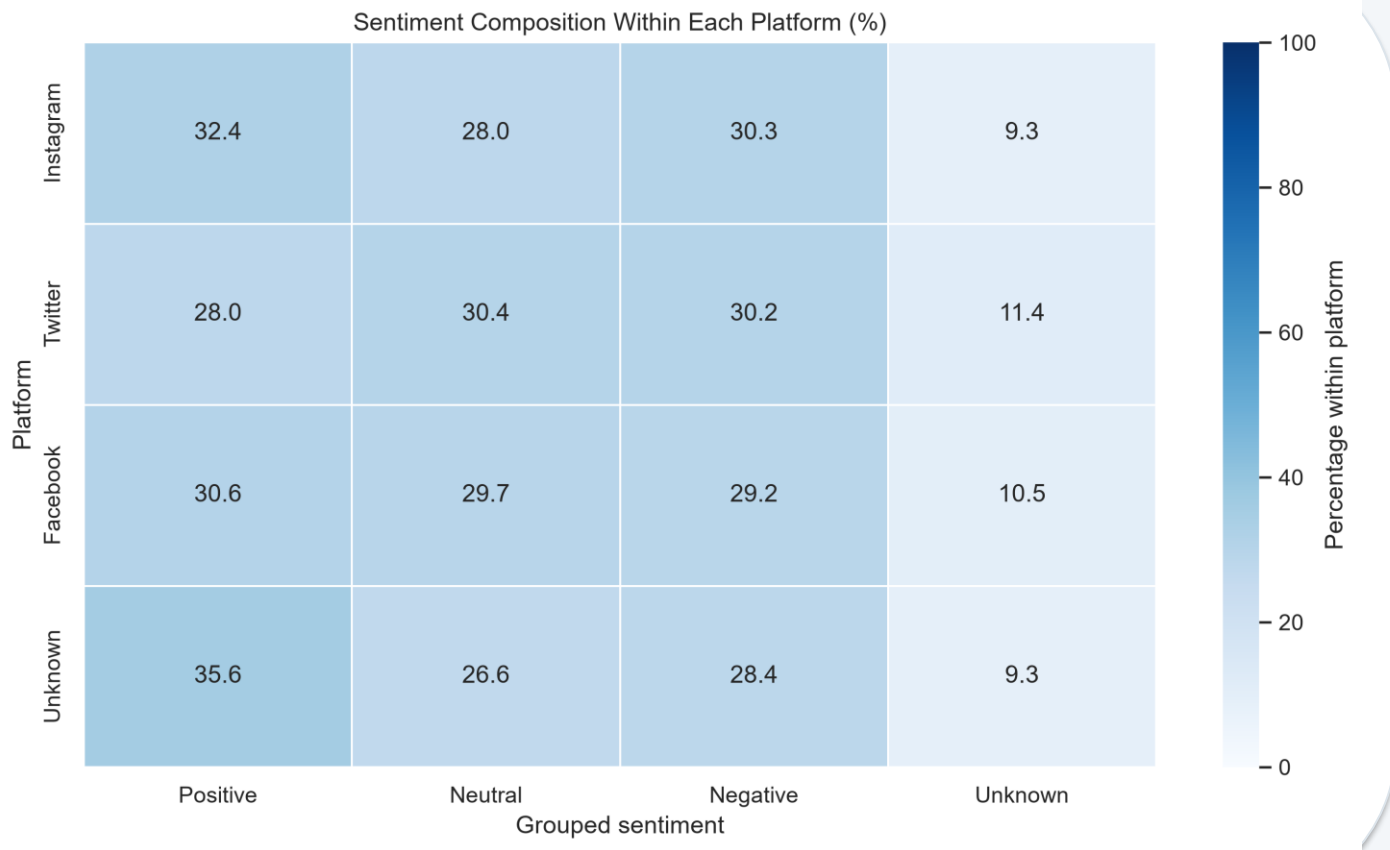
- Most records fall between July 2022 and June 2024.
- Within that interval, monthly counts range 103–147.
- May 2024 is highest (147); November 2023 is lowest (103).
- Eleven records sit outside the central period and matched source dates.

Do not interpret zero-count months or edge changes as real demand shifts.

Chart: 05_monthly_transaction_counts.png | Week 7 visualization workflow.

Sentiment shares are similar across named platforms

Percentages are calculated within platform across 2,980 interaction records.



WHAT THE DATA SHOWS

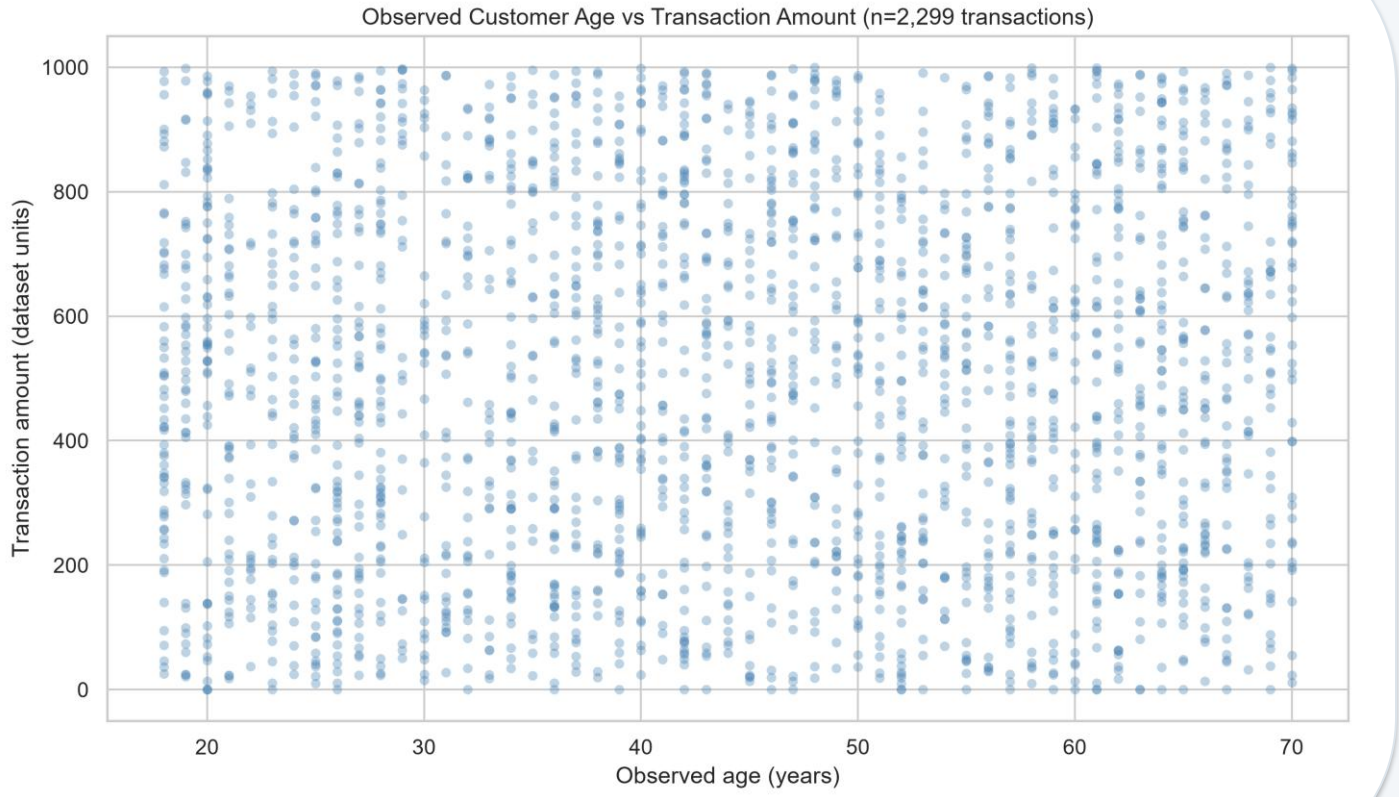
- Positive share: Instagram 32.4%, Facebook 30.6%, Twitter 28.0%.
- Positive, neutral, and negative shares are broadly similar.
- Unknown sentiment: 9.3%–11.4% across the named platforms.
- The Unknown platform group contains 289 records.

Keep unknown sentiment distinct; verify customer-to-social linkage before joining.

Chart: 06_sentiment_by_platform_heatmap.png | Week 7 visualization workflow.

Age alone shows little linear relationship with amount

2,299 transactions from 1,510 matched customers with observed age and valid amount.



WHAT THE DATA SHOWS

- Pearson correlation ≈ 0.0315.
- 27 valid-amount records had no demographic match.
- 316 matched records lacked an observed age.
- Repeated transactions mean points are not independent customers.

Do not use age as a standalone amount predictor; test any value in a controlled model.

Chart: 07_age_vs_transaction_amount.png | Week 7 visualization workflow.

Data limitations shaped what we could safely conclude

We preserved uncertain records for review, made assumptions visible, and avoided turning data-quality issues into false business findings.

CONFLICTING IDENTIFIERS	Repeated IDs can represent identity conflicts or legitimate repeated activity. We retained flagged records in cleaned data and audit exports, then excluded unresolved conflicts from descriptive analysis.
MIXED / INVALID AMOUNTS	The source included missing values, negative amounts, and text such as “Free.” We preserved zero values, did not impute invalid amounts into summaries, and flagged unresolved cases for business confirmation.
INCOMPLETE DATE COVERAGE	Dates span 2022–2024, but records are concentrated in a central window. Empty months and edge changes may reflect collection coverage, so we did not claim growth or seasonality.
CROSS-SOURCE MATCHING	Age–amount analysis matched 2,299 transactions from 1,510 customers; 27 valid-amount records lacked a demographic match and 316 matched records lacked observed age.
INTERPRETATION LIMITS	Small associations and visual similarities are descriptive, not proof of independence, lack of bias, or predictive value. A target, valid temporal split, and future-period evaluation are still needed.

Source: Week 3 preprocessing/EDA and Week 7 visualization findings; see the supporting workflow and audit exports.

PROPOSED ML SOLUTION

Pilot a time-aware purchase-propensity score

Recommendation only: no target is defined and no predictive model has been trained or evaluated.

1 | BUSINESS OUTCOME

Candidate target: at least one verified purchase in the 30 days after a customer as-of date. Confirm the outcome and action with FinMark first.

2 | CANDIDATE FEATURES

Only information available by the as-of date: purchase recency, frequency, spend, category mix, valid demographics, and—if linkage is approved—prior social aggregates.

3 | EVALUATION

Use chronological train/validation/test windows. Compare with a simple baseline; report precision/recall, PR-AUC, calibration, and business lift at an actionable threshold.

LEAKAGE-SAFE DESIGN

Separate feature lookback from the 30-day outcome window. Never use future transactions, post-outcome interactions, or test-period statistics in training.

GO / NO-GO CONDITIONS

Proceed only after transaction-date completeness, customer identity resolution, target feasibility, and permitted use of demographic/social data are signed off.

Recommendation based on available schemas and Week 3/7 findings; proposed target and evaluation are not yet validated.

RECOMMENDATION

Resolve the data and decision questions before building a model

The analysis is descriptive and useful for scoping; operational value must be demonstrated in a future pilot.

DECISION 1 | BUSINESS OWNER

Which customer action should a score prioritize—and what measurable outcome would make it valuable?

DECISION 2 | DATA STEWARDS

Confirm date conventions, periods of complete capture, identity conflicts, and approved cross-source joins.

DECISION 3 | MODEL GOVERNANCE

Approve target, eligible population, fairness checks, monitoring, and a human-reviewed pilot plan.

NEXT | EVIDENCE-BASED PILOT

Build a leakage-safe baseline; compare to a simple rule; validate calibration and lift on future periods before any deployment.

Recommendation: fund data validation and target design first; release a model only if temporal testing shows useful, fair, actionable lift.

Source: FinMark project data; Week 3 EDA and Week 7 visualization workflow.

Choose the least complex approach that answers the business question

The current evidence supports trustworthy descriptive analytics now; predictive modeling remains conditional on target and data validation.

A | EDA + DASHBOARD

Use now

Answers: What happened and where are the data gaps?

Pros: transparent, immediately supported by current work.
Trade-off: does not predict an individual outcome.

B | RFM / RULES

Benchmark

Recency-frequency-monetary summaries or a simple business rule.

Pros: interpretable baseline; low model complexity.
Trade-off: still needs valid dates and an agreed action.

C | PROPENSITY MODEL

Candidate next step

Predict a defined future purchase after an as-of date.

Pros: ranks customers for a chosen action.
Trade-off: requires reliable labels, dates, linkage, and prospective validation.

D | OTHER OPTIONS

Clustering: exploratory segments, not outcomes.

Time-series forecasting: requires complete, stable periods.

Neither is justified by current data coverage alone.

DECISION

Deliver A now; use B as the baseline and compare it with C only after stakeholder sign-off. If dates, target, or identity checks fail, stop at descriptive reporting rather than forcing a model.

Source: FinMark project data; Week 3 EDA and Week 7 visualization workflow.

Use methods that match the question—and report what has not been tested

The metrics below summarize current descriptive analysis; predictive performance is not yet available.

DISTRIBUTIONS

Histograms and robust summaries (median, range, spread) describe observed ages and valid amounts; imputed ages stay separate.

DATA ANOMALIES

Exact-duplicate checks and ID-conflict audits preserve questionable records for review. The within-category 1.5 × IQR check flagged no amount outliers; that is not proof of clean data.

ASSOCIATIONS

Cross-tabs and Cramér's V assess categorical association; Pearson $r \approx 0.0315$ describes a weak linear age–amount association in this matched transaction subset.

CANDIDATE CLASSIFIERS

If approved: logistic regression as an interpretable baseline; compare with a regularized tree-based challenger for nonlinear patterns. Select only on held-out evidence.

TEST DESIGN

Create customer as-of snapshots; use a chronological holdout that mirrors future scoring. Fit imputers, encoders, and scalers on training data only; prevent future-event leakage.

DECISION METRICS

Compare with the base rate and RFM rule. Report PR-AUC, precision/recall at campaign capacity, calibration (e.g., Brier score), and lift. Accuracy alone can mislead when classes are imbalanced.

No target, classifier, validation score, or claim of predictive accuracy exists yet. The proper current conclusion is “not evaluated,” not “high accuracy.”

Method references: NIST/SEMATECH EDA Handbook; scikit-learn evaluation and leakage guidance; Google ML Crash Course.

A traceable pipeline makes every cleaning decision reviewable

Keep source records intact, make transformations reproducible, and define a rollback before any pilot.

1 | FREEZE & PROFILE

Keep raw workbooks read-only. Record schema, row counts, date range, duplicate counts, and missingness by source.

2 | AUDIT & CLEAN

Write cleaned outputs separately. Preserve original values and flag imputations, invalids, and conflicting IDs; export exclusions with source row references.

3 | VERIFY EDA

Assert row-count reconciliation and subset rules; compare summaries to source files; record exclusions and unresolved assumptions.

4 | DEFINE THE OUTCOME

Agree the customer action, eligible population, 30-day label, as-of dates, feature window, and permitted data joins with business and data owners.

5 | TRAIN & CHALLENGE

Use temporal partitions and train-only preprocessing. Log data versions, code, parameters, metric trade-offs, subgroup checks, and model-versus-baseline results.

6 | PILOT & MONITOR

Start with a human-reviewed, limited pilot. Monitor calibration, lift, drift, coverage, and subgroup outcomes against agreed stop rules.

ROLLBACK: pause scoring and outreach; revert to the approved manual/RFM process; retain logs and raw data; correct the issue, rerun validation, and require sign-off before restart.

Governance principle: NIST AI RMF 1.0—validity, reliability, transparency, accountability, and managed risk.

Methods and safeguards are grounded in published guidance

Project-specific values and charts come from Team PETIX's Week 3 preprocessing/EDA and Week 7 visualization workflow.

1 | NIST/SEMATECH — Exploratory Data Analysis

Graphical EDA and exploratory methods.
https://www.itl.nist.gov/div898/handbook/eda/eda.htm

2 | NIST/SEMATECH — Detection of Outliers

Outlier screening supports investigation; statistical flags are not automatic proof of error.
https://www.itl.nist.gov/div898/handbook/eda/section3/eda35h.htm

3 | scikit-learn — Common Pitfalls

Consistent preprocessing and prevention of train/test leakage.
https://scikit-learn.org/stable/common_pitfalls.html

4 | scikit-learn — Cross-validation

Evaluation design; use time-aware splitting when observations are time ordered.
https://scikit-learn.org/stable/modules/cross_validation.html

5 | scikit-learn — Metrics and Scoring

Choose scores to match the prediction and decision task.
https://scikit-learn.org/stable/modules/model_evaluation.html

6 | Google ML Crash Course — Classification Metrics

Precision, recall, and accuracy depend on class balance and error costs.
https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

7 | NIST AI Risk Management Framework (AI RMF 1.0): https://www.nist.gov/itl/ai-risk-management-framework

References accessed 27 Oct 2026. Project results are from the supplied Week 3 and Week 7 materials.